

Contact Modes Are Strata

What Geometric Structure Buys in Discrete–Continuous Planning

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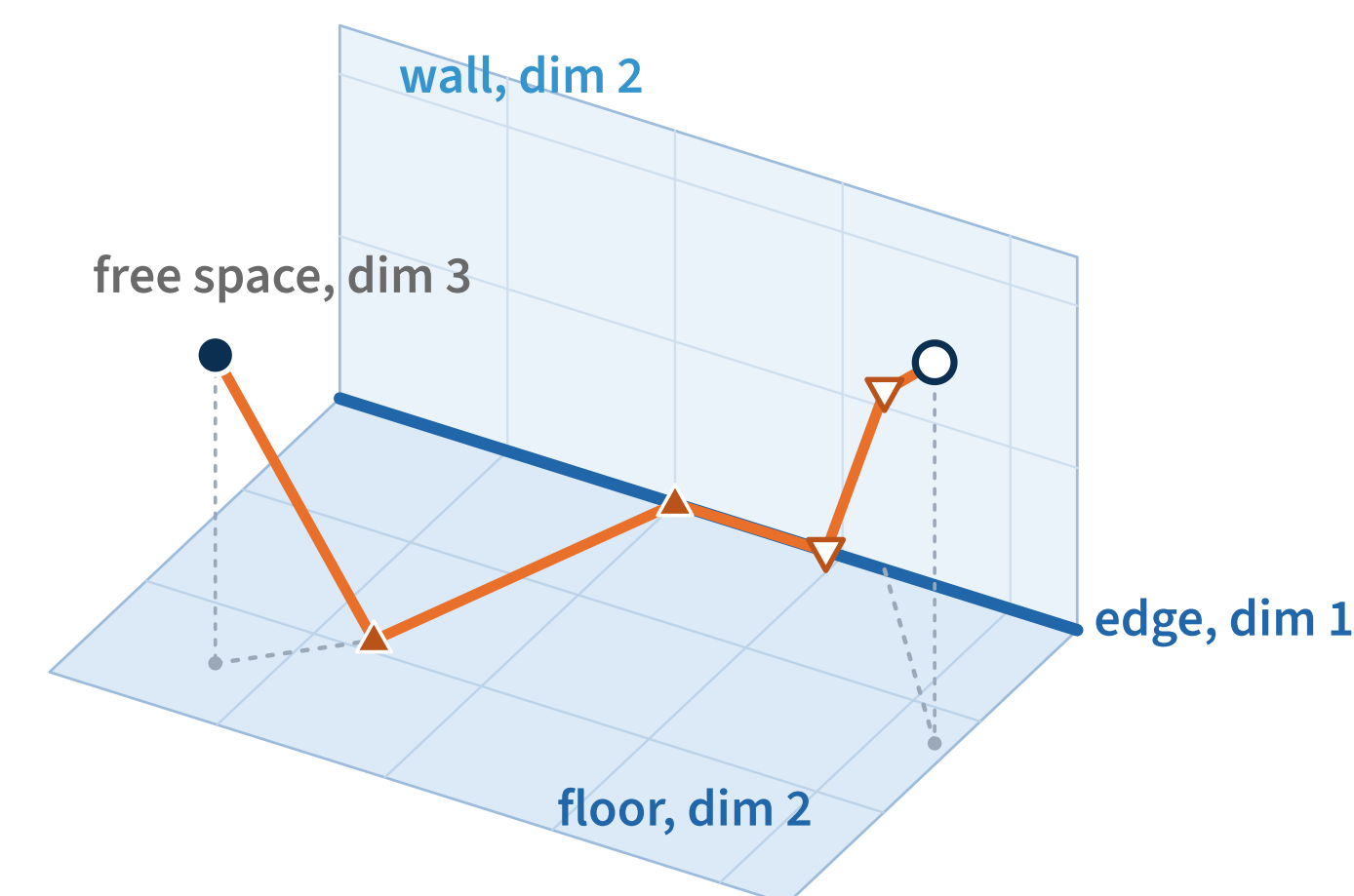
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A contact mode **is** a stratum of the configuration space.

A plan is a **walk over strata**, following a geodesic within each one.

We do not provide the planner with any mode, contact sequence, or manifold. The strata and the transitions between them follow from the signed distances and gradients a collision engine already provides.



1 What planners are given

Frameworks for discrete–continuous planning take the **discrete part as input**.

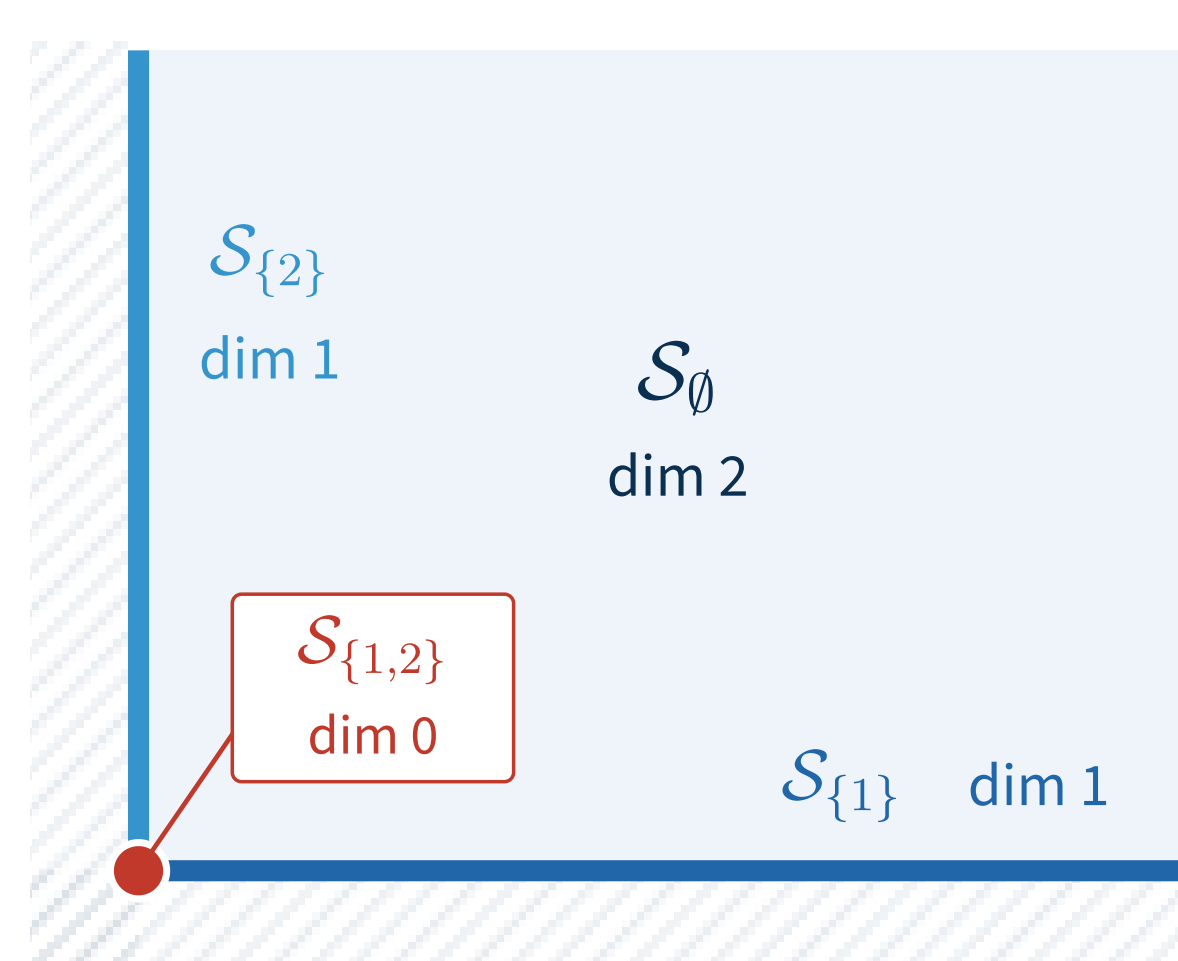
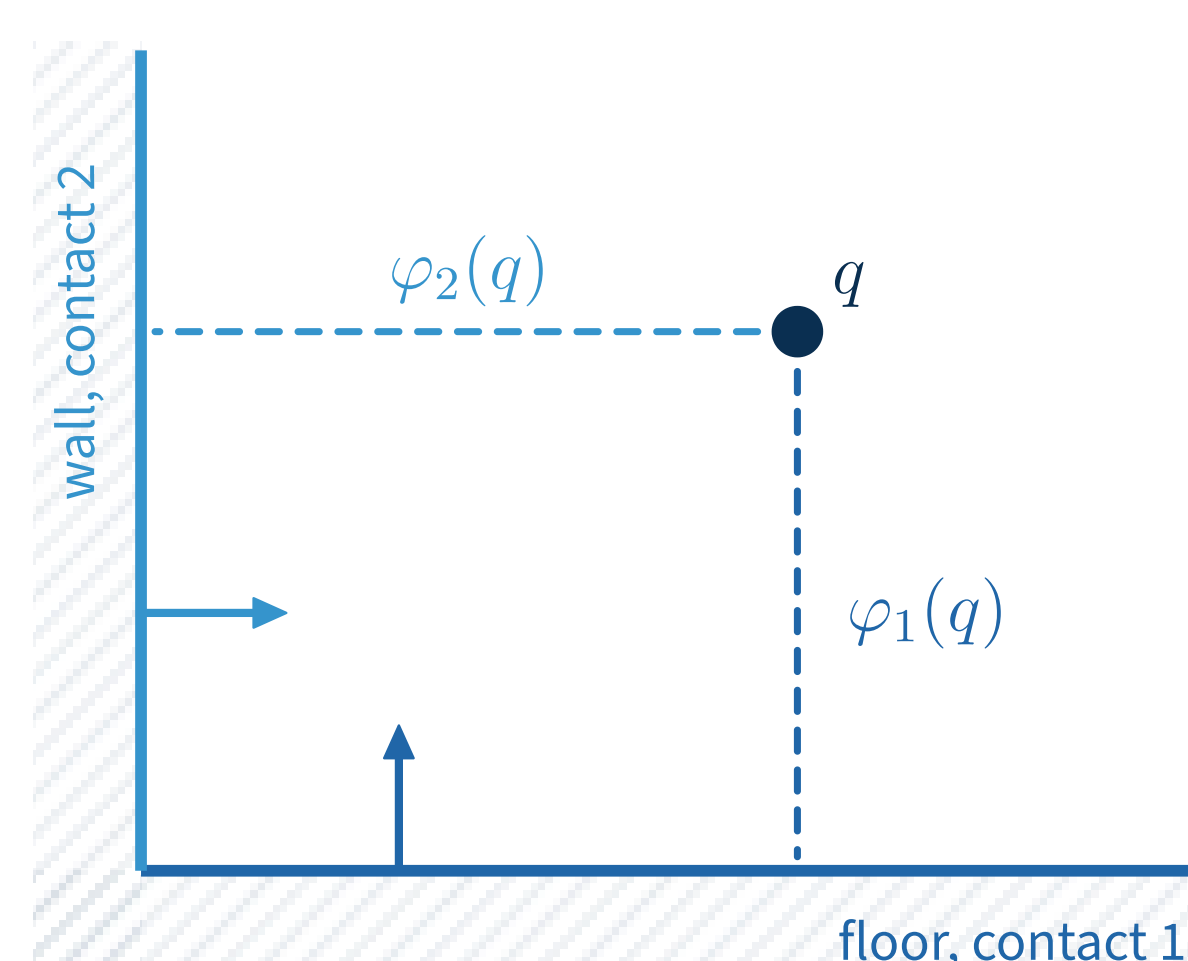
FRAMEWORK	NEEDS AS INPUT
Graph of Convex Sets Marcucci et al. 2024	a convex decomposition
Logic-Geometric Programming Toussaint 2015	a symbolic domain
Multi-Modal Motion Planning Hauser & Latombe 2010	a mode family
Sequenced Manifolds Englert et al. 2020	the manifold sequence itself
This work	signed distances and gradients, from a collision engine

PAIRS THAT COULD TOUCH	CONTACT MODES
4 fingertips	16
10	1,024
240 the hand	$2^{240} \approx 10^{72}$

Most of these modes are infeasible, since many contacts cannot occur simultaneously, and finding the feasible ones is itself a planning problem.

2 Every contact removes a dimension

Group configurations by **which signed distances vanish**. Each group is a manifold, and every contact it holds removes one dimension.

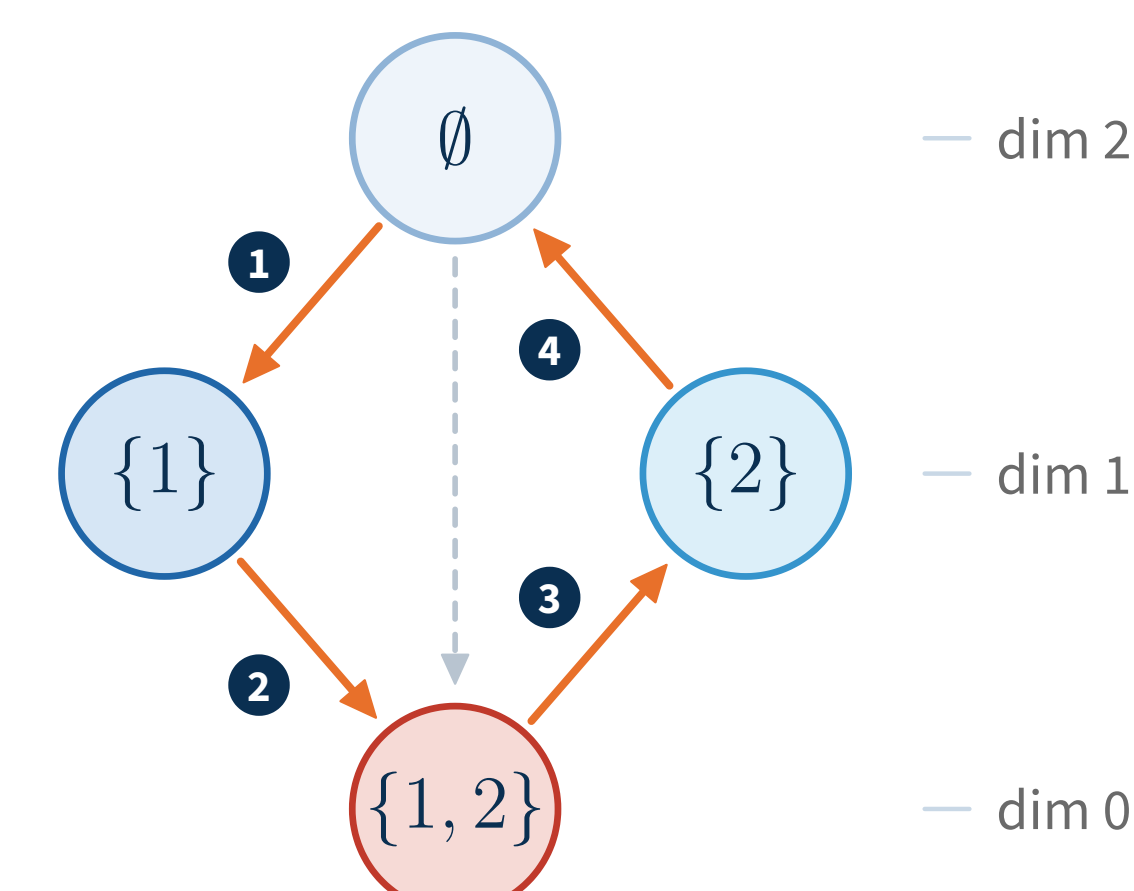
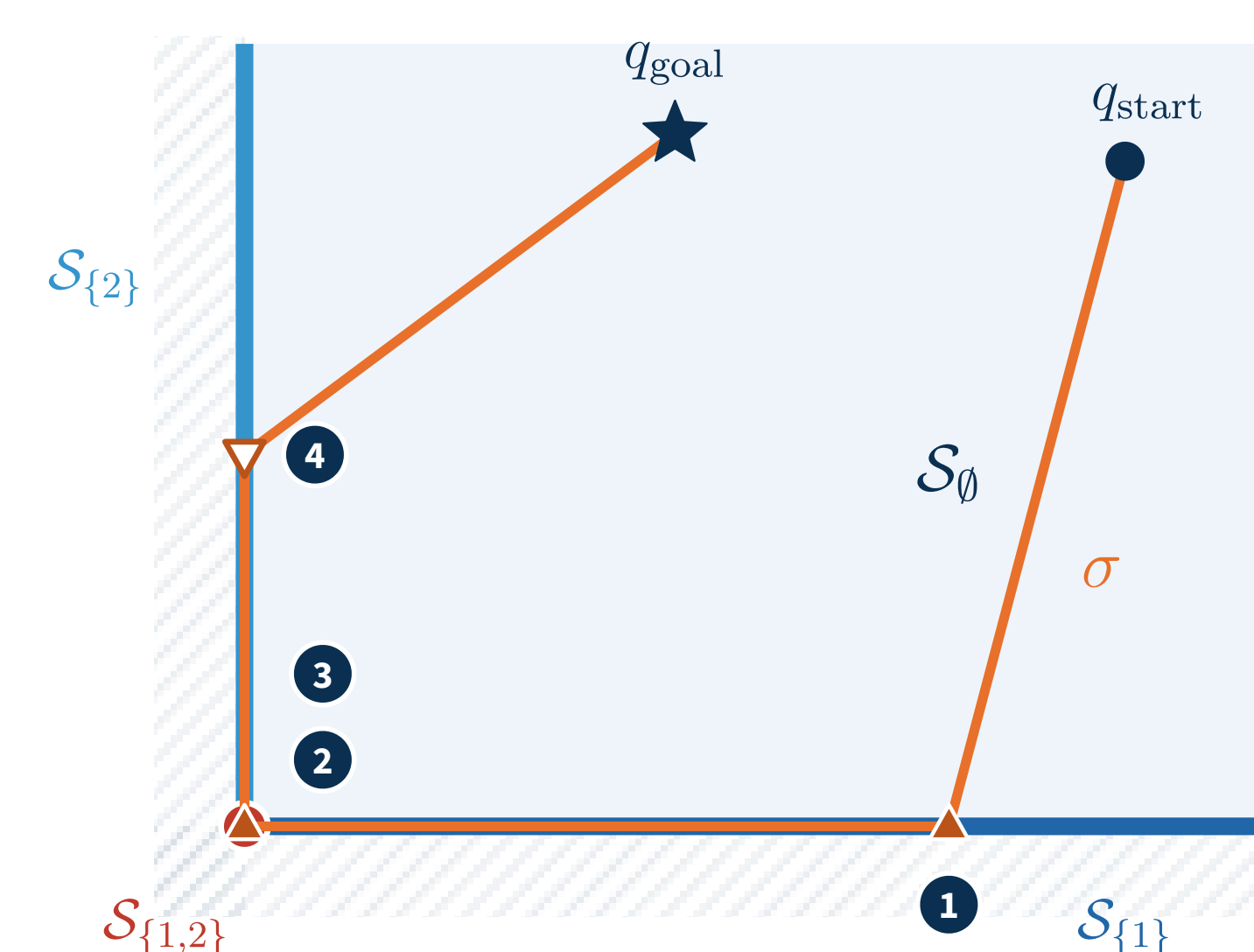


$$\mathcal{S}_A = \{q \mid \varphi_i(q) = 0 \ \forall i \in A, \ \varphi_j(q) > 0 \ \forall j \notin A\} \quad \dim \mathcal{S}_A = n - |A|, \quad \mathcal{T}_q \mathcal{S}_A = \ker J_A(q)$$

A configuration space glued together from manifolds of different dimensions is a **stratified space**, and its manifolds are its **strata**.

3 A plan is a walk over strata

Making contacts steps **down** the stratum graph and breaking them steps back **up**. Within each stratum the walk follows a geodesic.

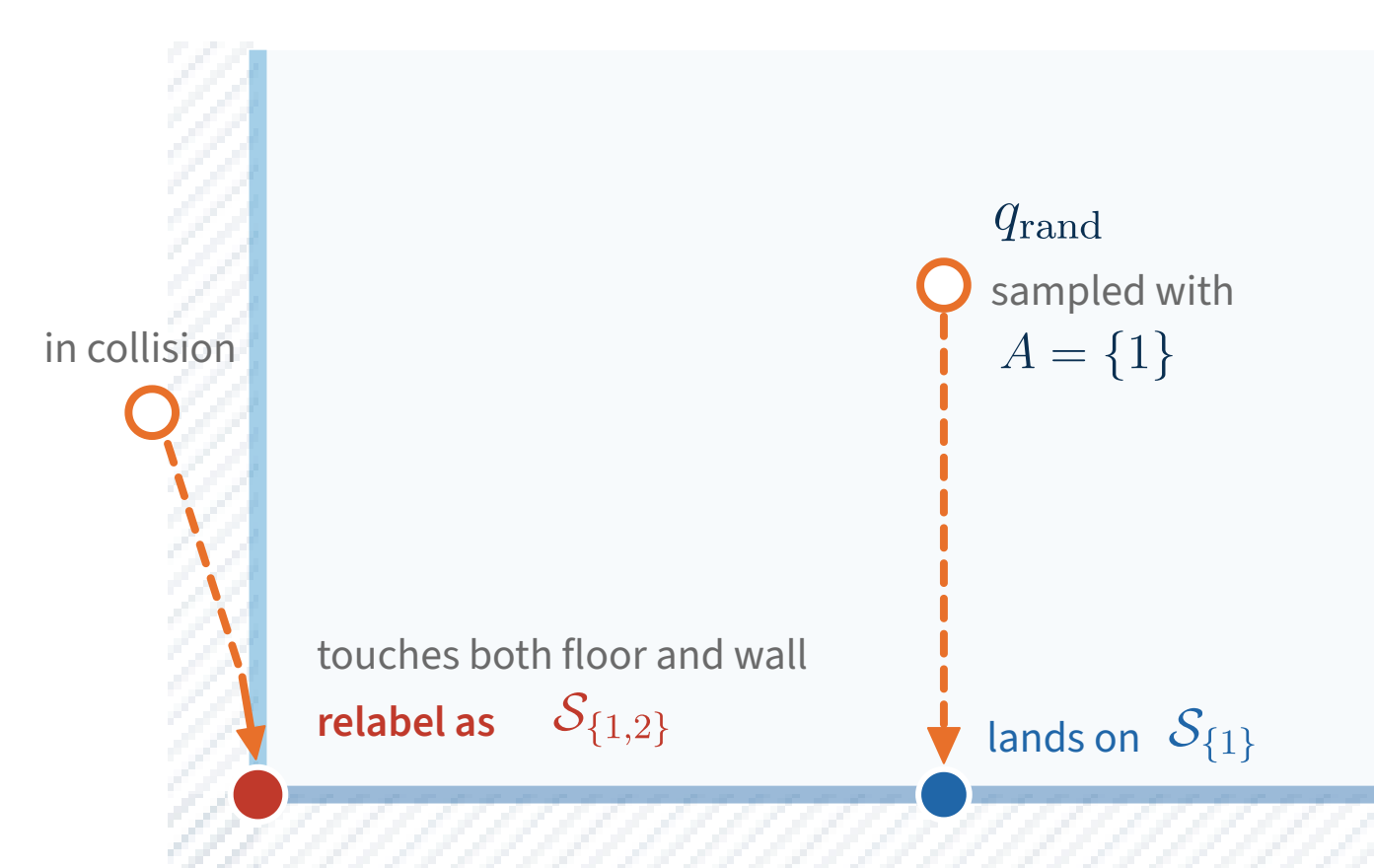


The planner never builds the stratum graph explicitly. Motion creates its edges implicitly, and the dashed edge makes both contacts at once.

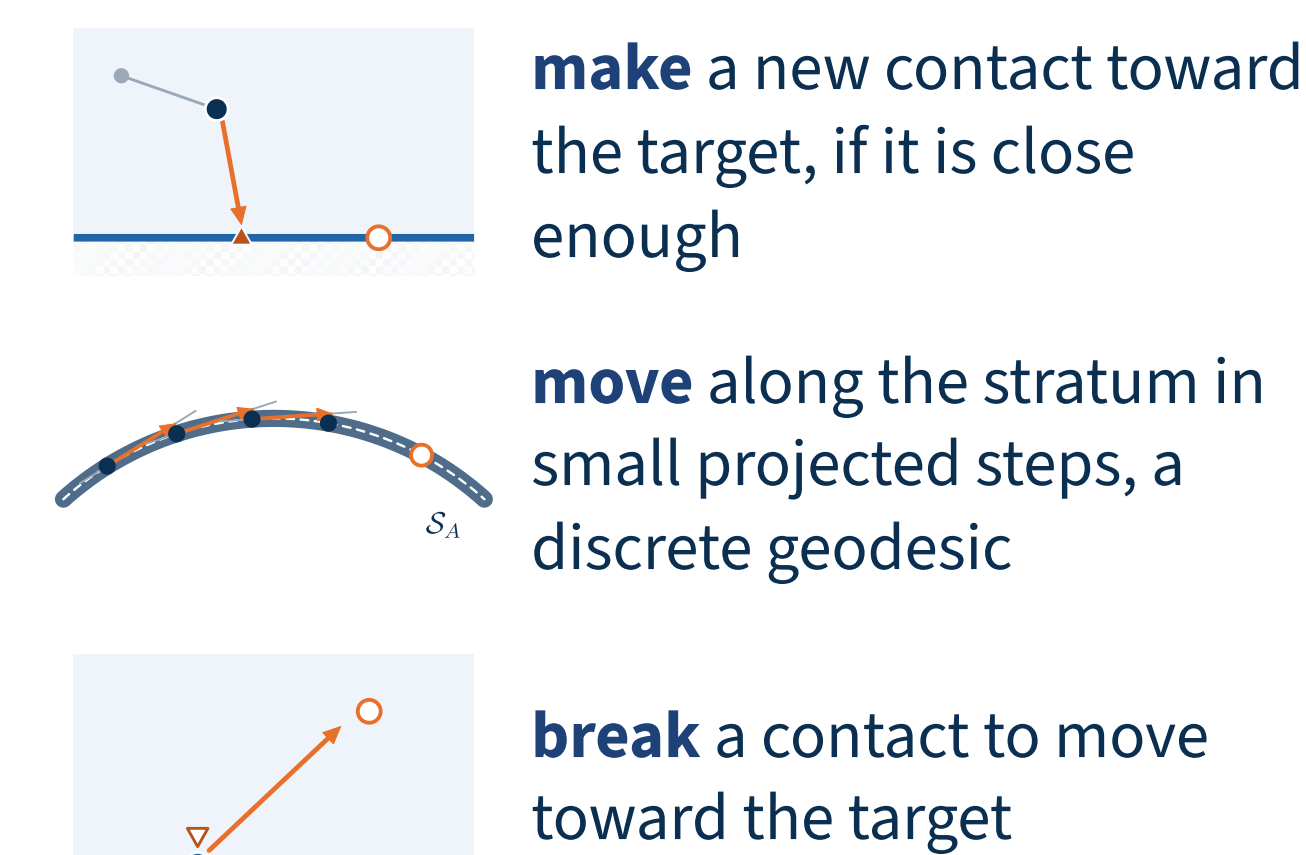
4 Planning on strata

A single-tree **RRT** grows over the strata. Lower-dimensional strata have measure zero, so we cannot use rejection sampling. Instead, we sample a contact set and an ambient configuration, then **project** the configuration onto that stratum.

sample, project, relabel



extend by make, move, break

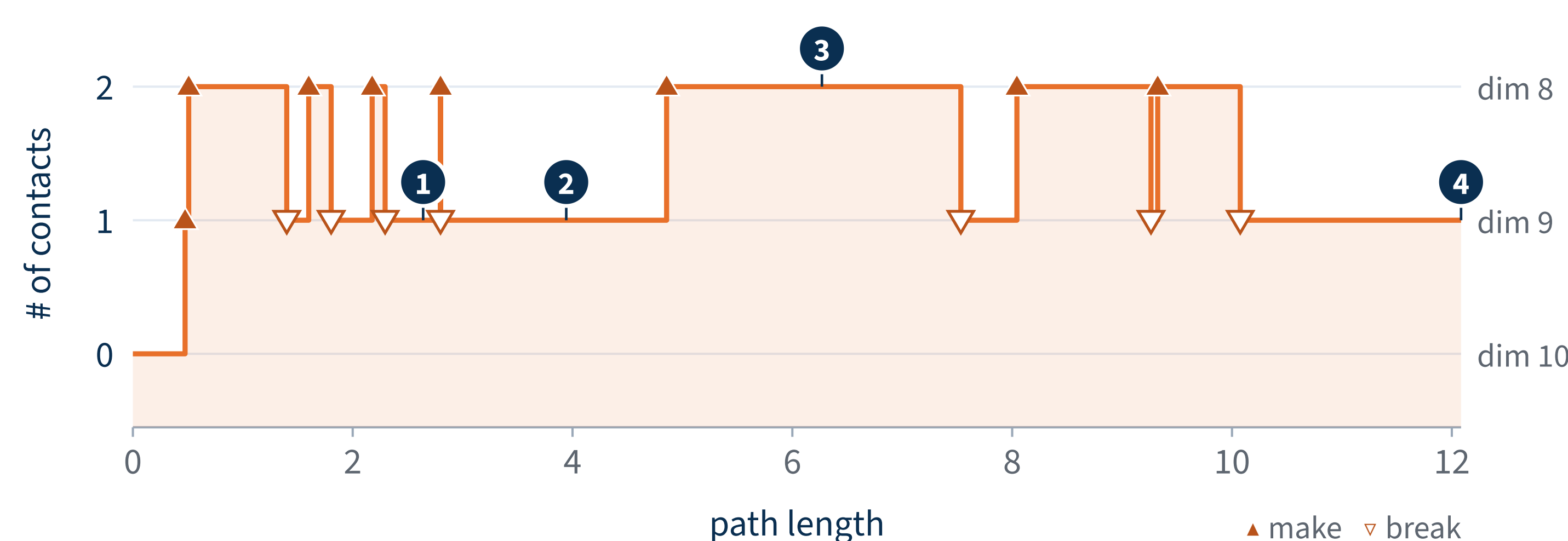
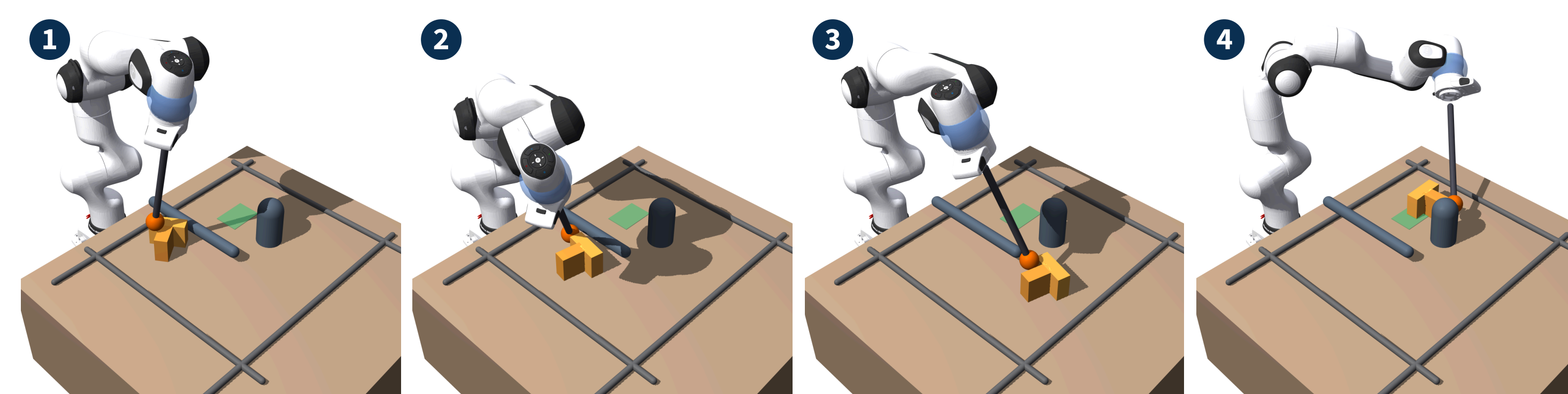


$$q \leftarrow q - J_A(q)^+ h_A(q), \quad h_A(q) = (\varphi_i(q))_{i \in A}$$

Relabeling allows the planner to **discover strata automatically**, and by construction every path in the tree is a hybrid path (a **discrete** sequence of strata with **continuous** motion within each).

5 Two contact-rich tasks, no modes given

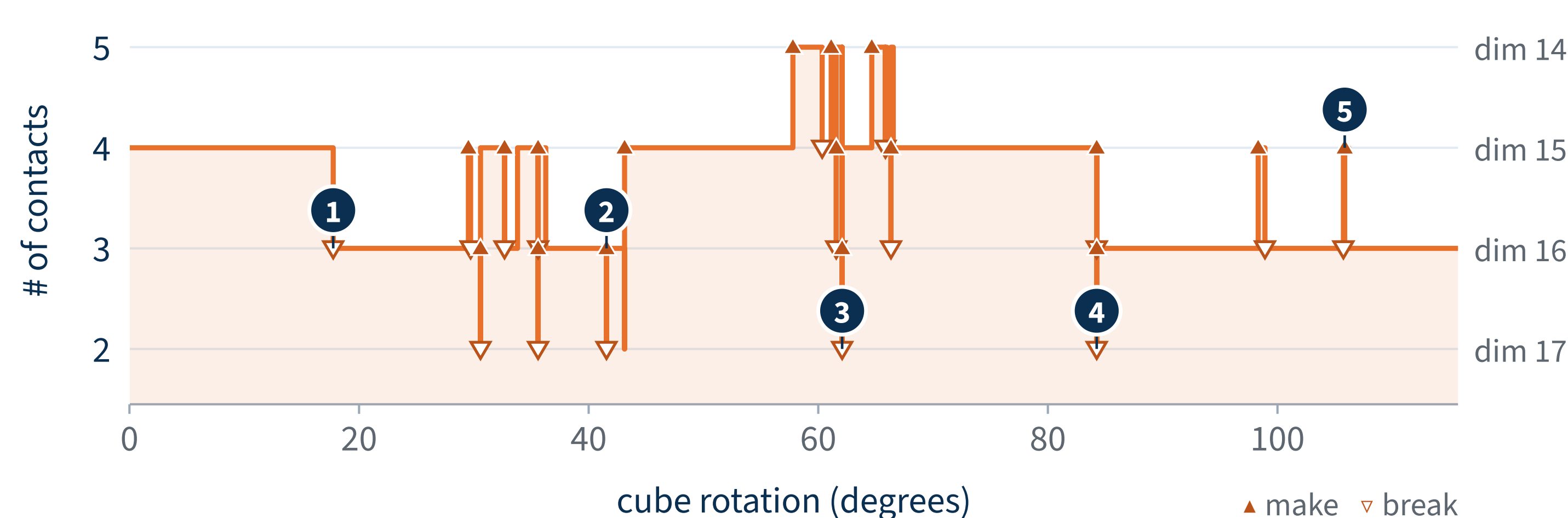
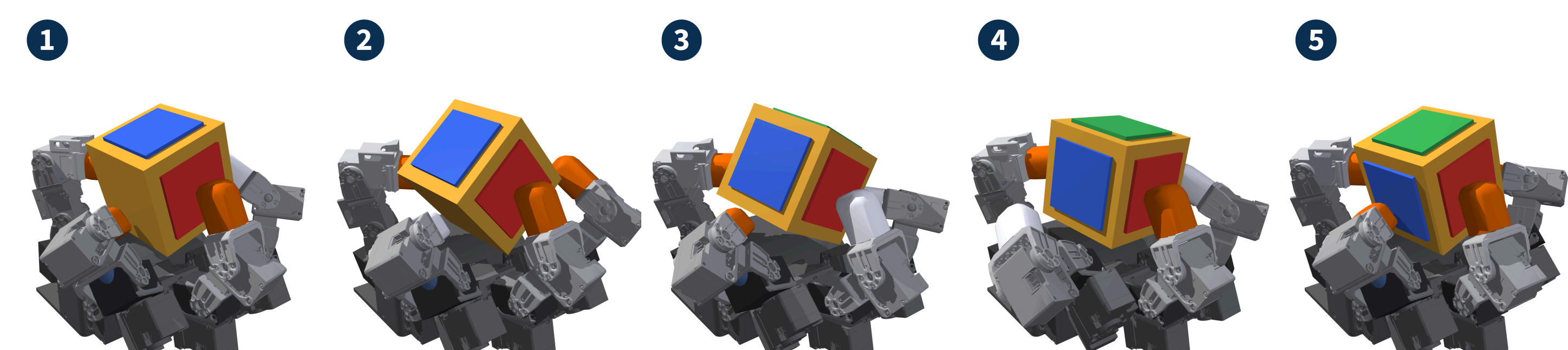
Franka FR3 pushes a T-shaped block · $n = 10$, 22 body pairs



760 ms
median planning time

5 / 4
median make / break

LEAP hand turns a cube toward 120° · $n = 19$, 240 body pairs



9.9 s
median planning time

52 / 52
median make / break